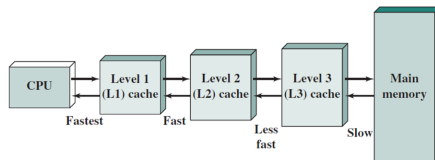


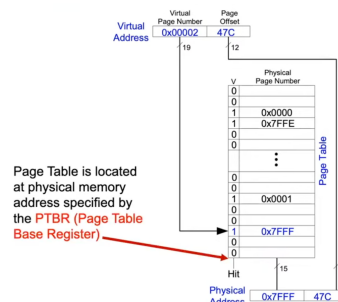
Week 6-8 cache

Topic Overview

Cache Memory.



Virtual Memory



DRAM vs SRAM	
FEATURE	DRAM (Dynamic RAM)
Technology Used	Uses 1 transistor + 1 capacitor per bit Word Line (WL) Bit Line (BL) 1 Transistor 1 Capacitor Stores data as charge in a capacitor
Working Principle	Stores data as charge in capacitors (needs refresh)
Storage Capacity (Density)	High density → large memory (GBs)
Typical Use	Main memory (RAM in computer)
Speed	Slower access time
Price (Cost per GB)	Low cost (~\$20 - \$70 per GB)
Power / Refresh	Requires periodic refresh Read → Refresh (Power) → Repeat
Application Focus	Cost-effective large storage
FEATURE	SRAM (Static RAM)
Technology Used	Uses 6 transistors (6T1R circuit) per bit Word Line (WL) Bit Line (BL) 6 Transistors 1 Capacitor Stores data using stable flip-flop state (no refresh needed)
Working Principle	Stores data using stable flip-flop state (no refresh needed)
Storage Capacity (Density)	Low density → small memory (MBs)
Typical Use	Cache memory (CPU L1, L2, L3)
Speed	Very fast access time
Price (Cost per GB)	Very expensive (~\$5000 per GB)
Power / Refresh	No refresh required
Application Focus	High-speed performance

This module introduces the basic concepts of the Central Processing Unit (CPU) memory hierarchy (caches). We will talk about why we have a hierarchy in the CPU memory, how is it implemented, and different types of caches to improve execution time. This unit is three-week in length, and you should expect to spend 8-12 hours per week completing the readings and activities.

Introduction to Cache and Memory Hierarchy

Learning Outcomes

By the end of this week you will be able to:

After the successful completion of this learning unit, you will be able to...:

1. Define the concept of cache as storage(memory) with locality (temporal and spatial)
2. Compare different types of caches and their influence on the CPU execution time
3. Develop a cache hierarchy for the pipelined CPU we implemented on the previous unit

The above Learning Objectives above align with the Course Learning Objective (list here):

1. To efficiently use memory hierarchies, including caches
2. To understand virtual memory

Concepts covered:

1. Cache
2. locality
3. Memory hierarchy
4. Types of Cache: Direct Mapped, N-way set associative and Fully Associative
5. Virtual Memory
6. Paging
7. Page Table and TLBs

Course Materials

The following materials will help you meet the learning objectives listed above

1. PowerPoint slides, that can be found here in pdf
 1. slides **6. Memory.pdf** (<https://canvas.calpoly.edu/courses/186931/files/21847976?wrap=1>)  (https://canvas.calpoly.edu/courses/186931/files/21847976/download?download_frd=1)
2. The book chapter "Chapter 5. CPU Memory. Cache" can be found here.
 1. **COD ch 5.1-5.3.pdf** (<https://canvas.calpoly.edu/courses/186931/files/21847977?wrap=1>)  (https://canvas.calpoly.edu/courses/186931/files/21847977/download?download_frd=1)
 2. **COD ch 5.7.pdf** (<https://canvas.calpoly.edu/courses/186931/files/21847973?wrap=1>)  (https://canvas.calpoly.edu/courses/186931/files/21847973/download?download_frd=1)
3. Practice hw book chapter and
4. Go to the following webpage: **Dinero V** (<https://github.com/danluu/dineroV/blob/master/README>) or this one **ParaCache** (<https://www3.ntu.edu.sg/home/smitha/ParaCache/Paracache/fa.html>), a cache visual simulator where you can change/design your own parameters for the cache and create visualizations for the number of memory hits and misses of your design. **Cache Hw practice** (<https://canvas.calpoly.edu/courses/186931/pages/cache-hw-practice?wrap=1>)

Activities and Formative Assessment

The following activities will help you master the learning objectives listed above:

1. **(Student interaction learning activity):** "Discussion: Block Size Considerations." Instructions for students: Locate on the web (article, stackoverflow comment, Wikipedia entry etc) information about how to best select the block size of a cache. Write a discussion post to describe the entry and if you agree or disagree with it and why. After posting, select one of your peers' posts, and view their description. Do you agree with the interpretation? Does your own interpretation differ? Discussion **Block Size Considerations** (https://canvas.calpoly.edu/courses/186931/discussion_topics/898829?wrap=1)
2. **(Reflective Analysis) :** Written Assignment. After attending the class and reviewing the week's materials (webpage, pdf file, and book chapter) submit a written reflection of what you have learned. Include the following: How would you describe what a cache is to a friend or peer? Why is a cache fundamental for CPU execution time improvement? Create your own cache hierarchy diagrams to explain the basic concept of the cache. What type of cache is your favorite cache? Explain why you think it is more efficient. Is there any concept you are still confused about? What else would you like to know? Discussion **Cache Concepts** (https://canvas.calpoly.edu/courses/186931/discussion_topics/898828?wrap=1)
3. **(Formative assessment):**
 1. Please complete the **Quiz: Cache Basics**. This quiz will review basic concepts.
 1. **Basic Quiz Cache (Index, Tag) Week 5** (\$CANVAS_OBJECT_REFERENCE\$/assignments/g6faef45d7cbb539bba8cec431c33b91c?wrap=1)
 2. **Quiz Cache and Virtual Memory Quiz** (\$CANVAS_OBJECT_REFERENCE\$/assignments/g1abc6e852689ddb5bf163a75b13e10e9?wrap=1)
 3. **Quiz Set Associative Cache** (<https://canvas.calpoly.edu/courses/186931/assignments/1612097?wrap=1>)
 2. Please complete the **Lab**:
 1. **Develop a Direct Map Cache for IM.** **lab 4 Direct Map** (<https://canvas.calpoly.edu/courses/186931/assignments/1612102>)
 2. **Develop a set assoc cache Cache for DM.** Simulation as described LAB Cache. **Lab 5: Caches** (<https://canvas.calpoly.edu/courses/186931/assignments/1612089?wrap=1>)